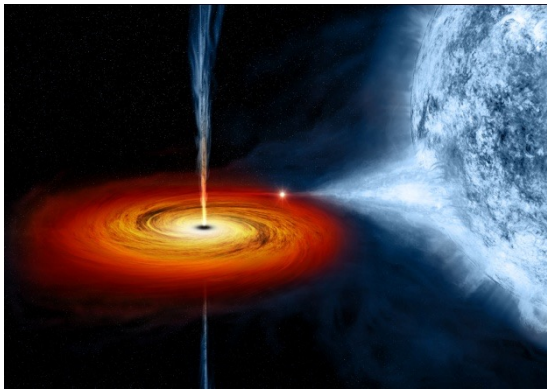
Master in Astrophysics, Particle Physics, and Cosmology
Academic year 2025-2026

High-energy astrophysics





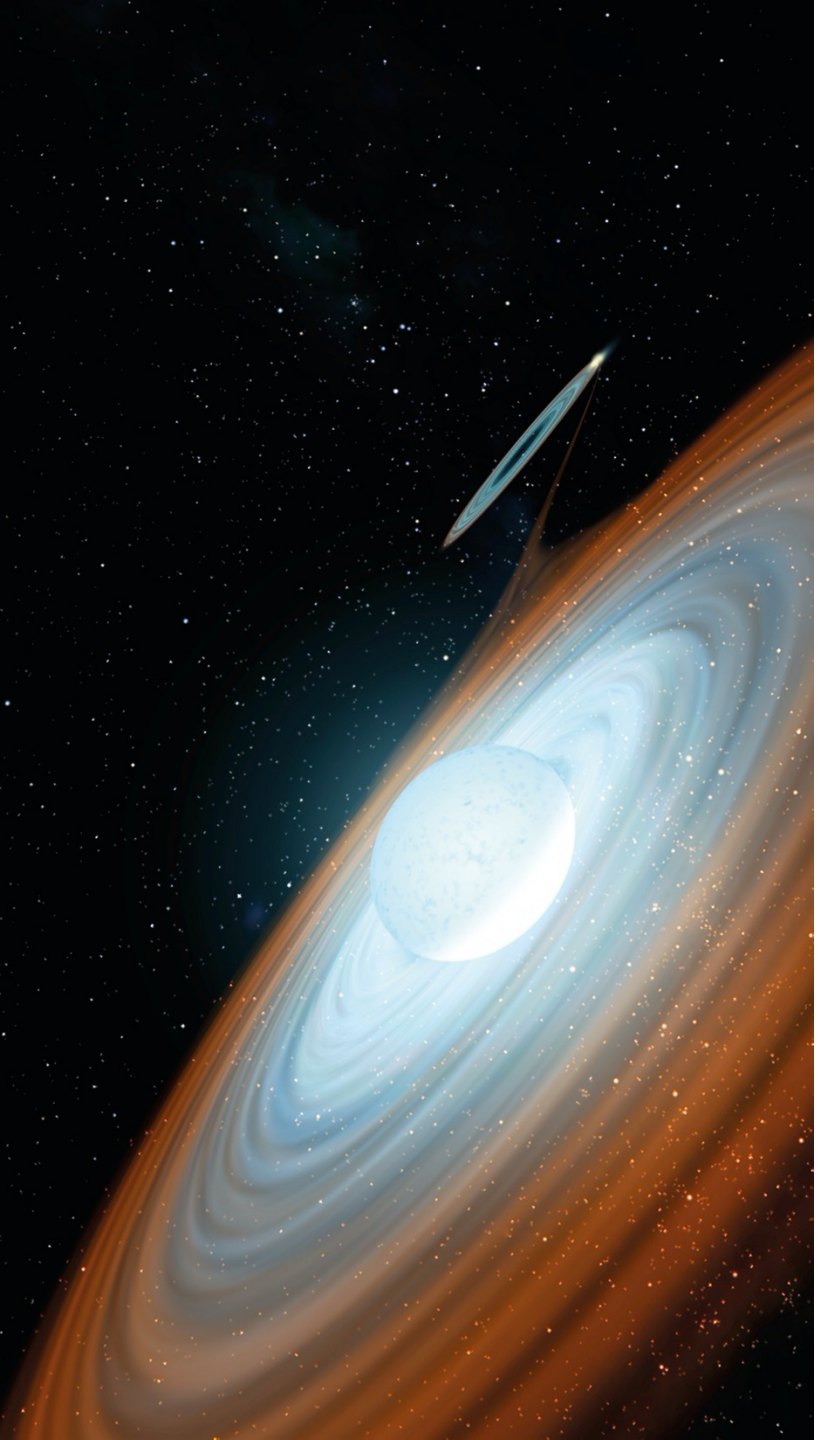
Valentí Bosch (vbosch@ub.edu)
Marc Ribó (mribo@ub.edu)

(https://icc.ub.edu/research/high-energy-astrophysics)

UNIVERSITAT DE BARCELONA

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Goals:

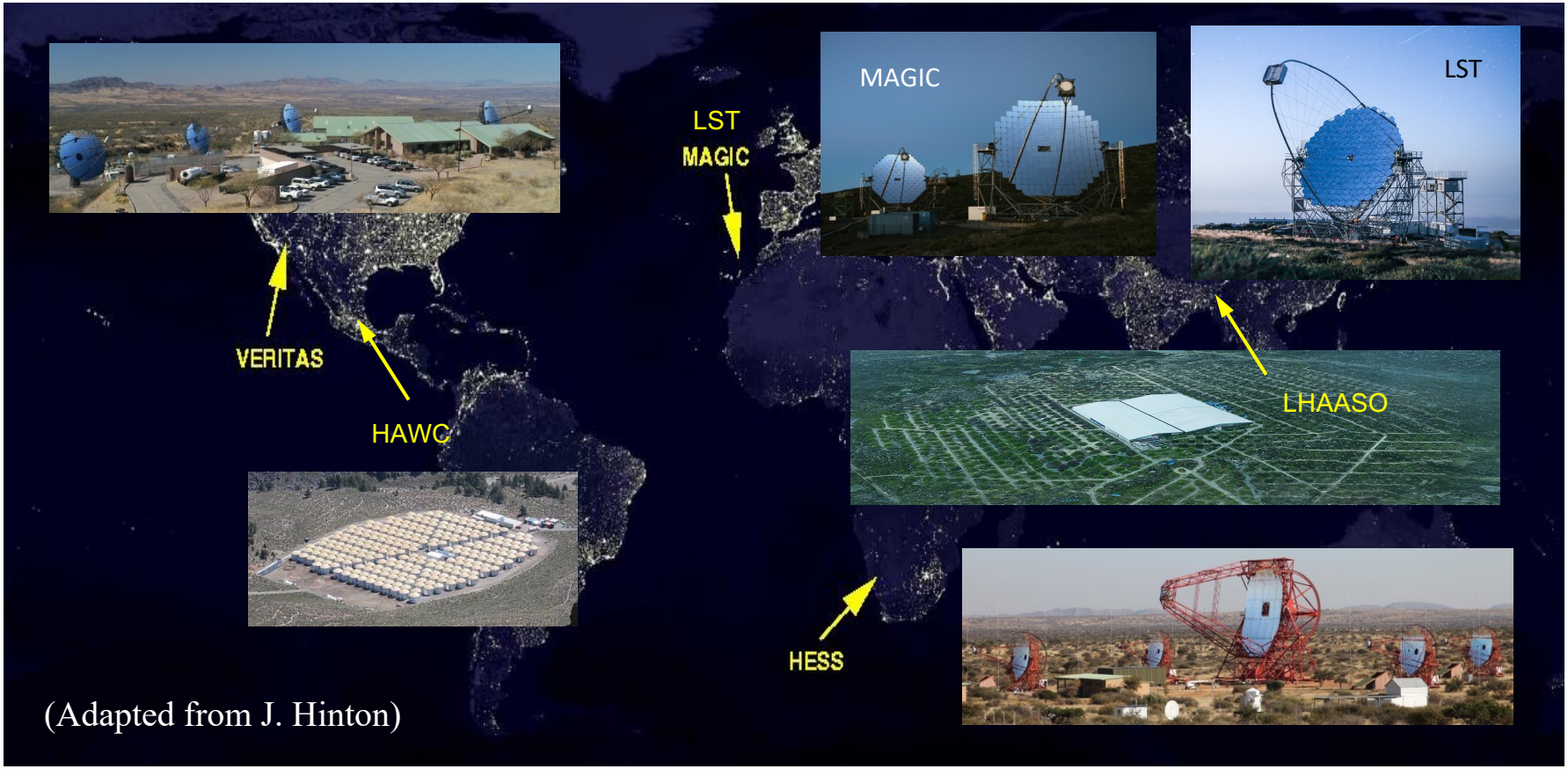
To train, from the observational and theoretical point of view, a group of future researchers in HE astrophysics.

To understand the:

- physical mechanisms capable of **accelerating particles** to high energies and associated **radiative processes**
- **phenomenology** of various kinds of HE astrophysical sources such as supermassive BH in galactic nuclei, XRB stars, pulsars, SNR
- most recent **observational results** and their impact in the models available

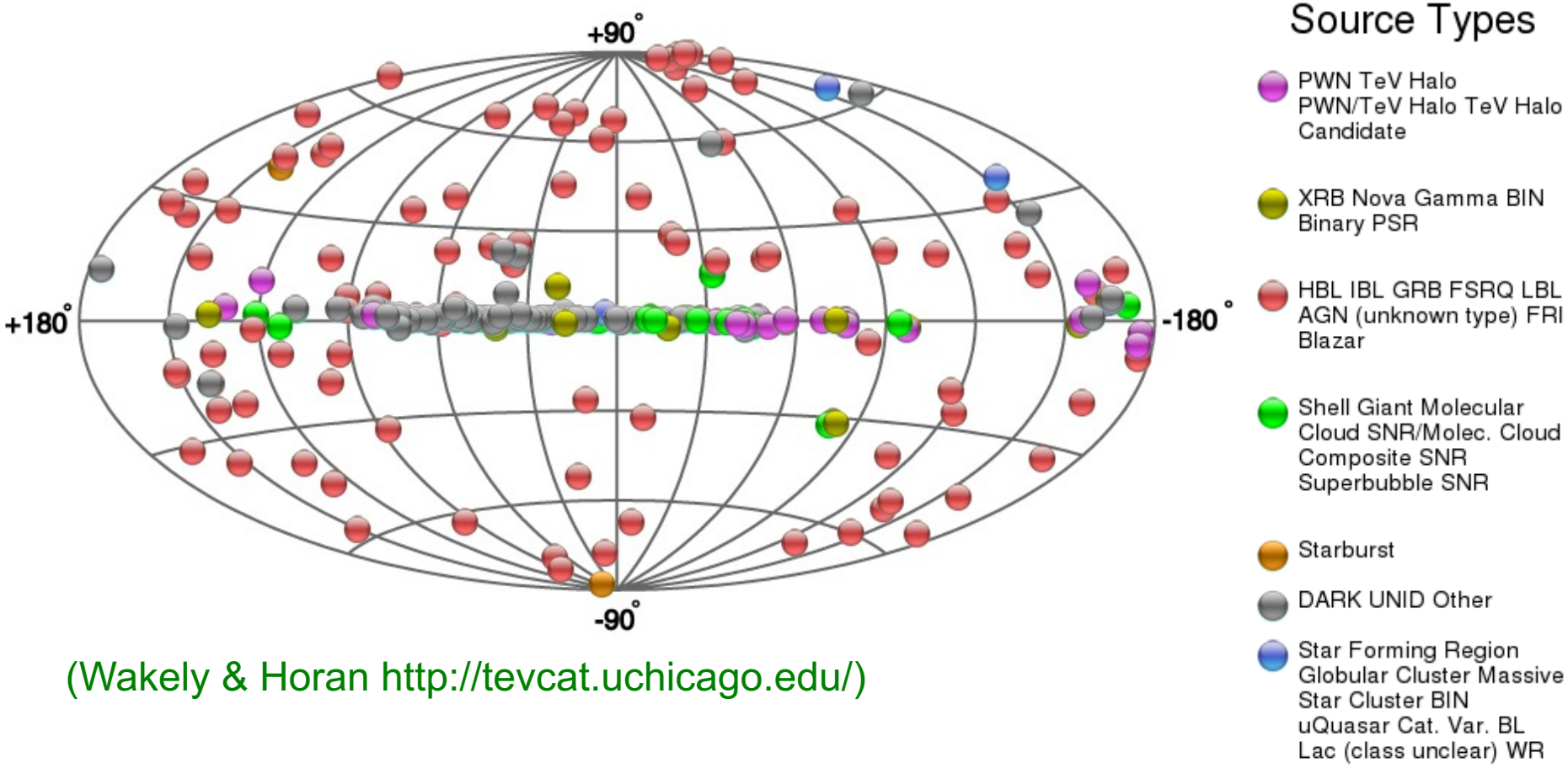
Current generation of TeV facilities:

- ~20 years of data from HESS, MAGIC and VERITAS.
- 8 years of data by HAWC.
- 4 yr of CTA/LST data → Soon much better sensitivity than previous facilities.
- 4 yr of data from LHAASO → PeVatrons everywhere!



TeV sources known

2025 Sep: 316 sources known! ~35% extragalactic, ~35% galactic, ~30% unid

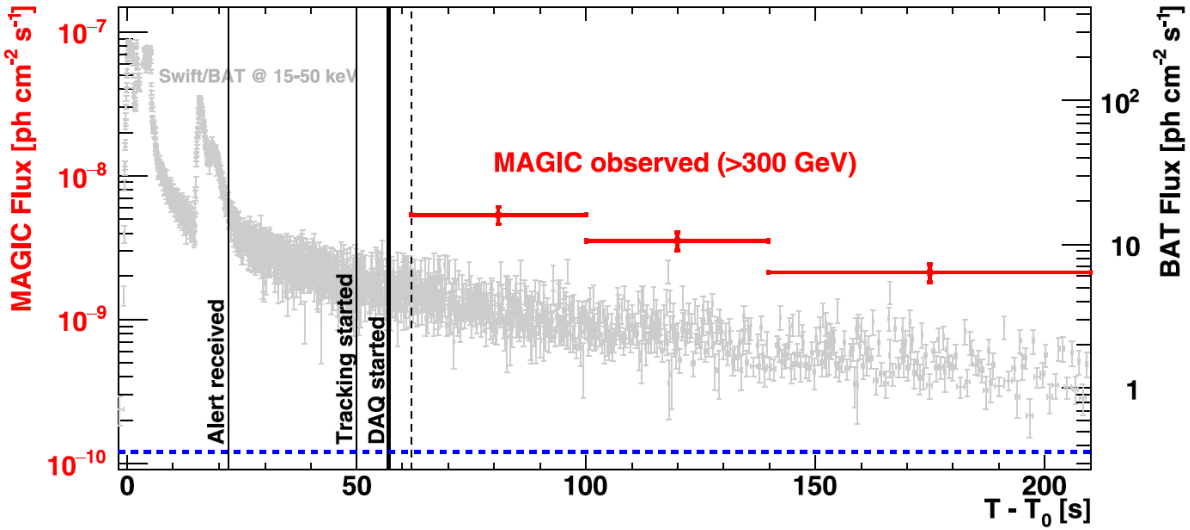


(Wakely & Horan http://tevcat.uchicago.edu/)

We know a similar number of source classes at VHE γ-rays and at HE γ-rays.

Gamma-Ray Bursts (GRBs).

MAGIC discovers GRBs at VHE: GRB 190114C.



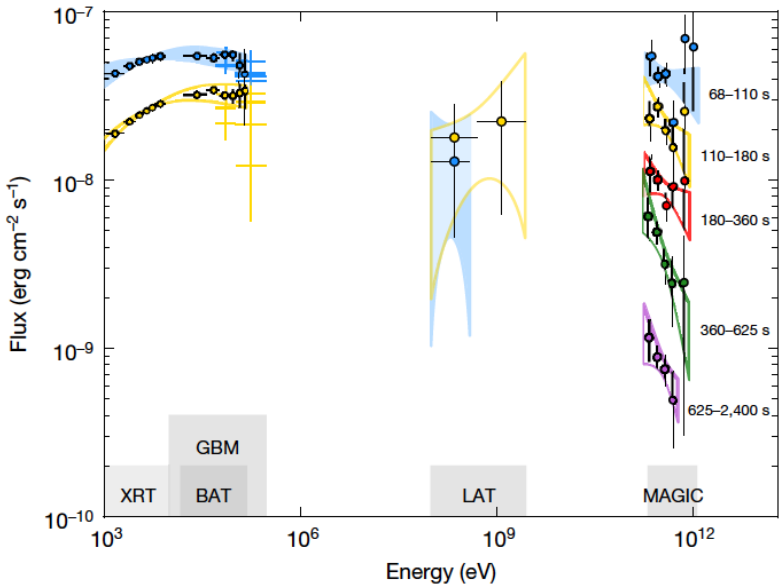
First GRB (long) reported at TeV energies.

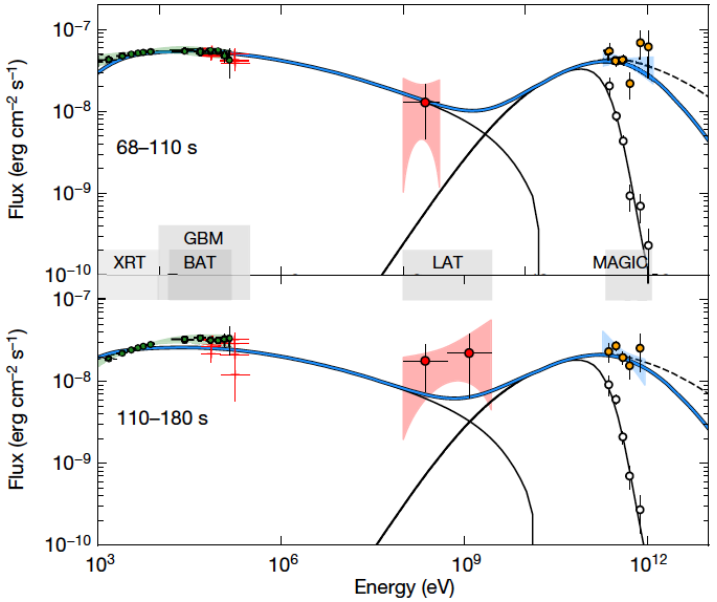
Redshift: z=0.425.

Detected **above 300 GeV**.

New IC component.

(MAGIC Collaboration 2019a & MAGIC Collaboration et al. 2019b).

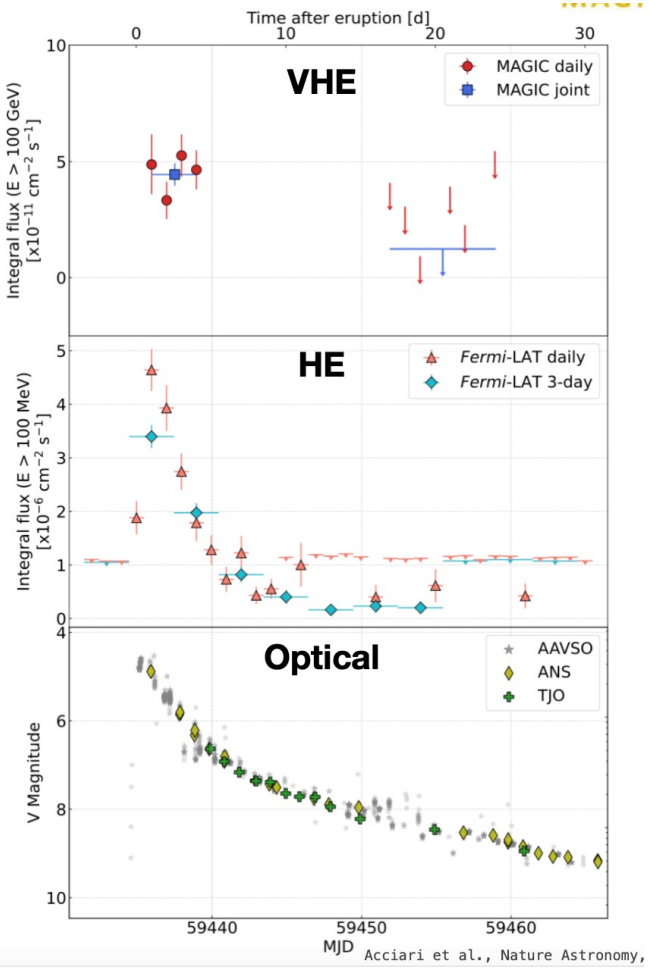


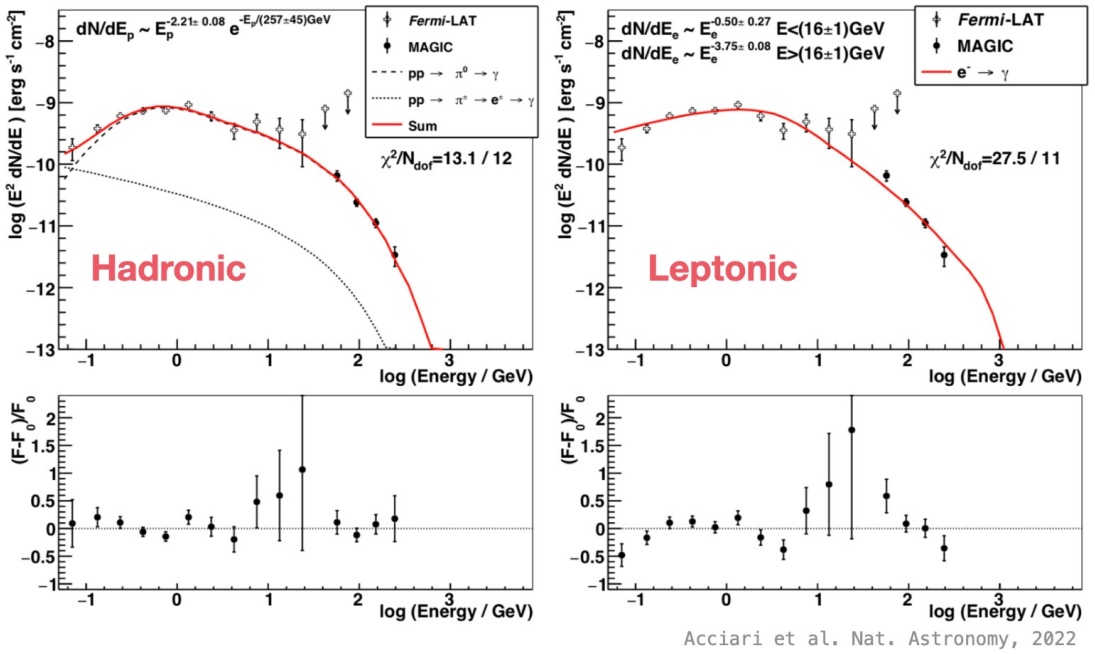


Novae.

MAGIC discovers proton acceleration in the recurrent nova RS Oph. (MAGIC Collaboration 2022).







Acciari et al. Nat. Astronomy, 2022

Different arguments favor **proton acceleration**:

- Flux evolution at different wavelengths.
- Spectral fitting at different epochs.
- Acceleration and cooling timescales.

Program: 26 h: 24 h lectures, 1 h exam, 1 h presentation

MR, 3 sessions
9 – 11 Feb



1. **Introduction**
    1. Cosmic rays
    2. Neutrinos
    3. Gravitational waves
    4. Electromagnetic waves
    5. Overview of stellar evolution and compact objects

2. **Particle acceleration and radiation mechanisms in high energy astrophysics**
    1. Particle acceleration
    2. Diffusion
    3. Energy losses
    4. Radiative processes (Thermal emission, synchrotron radiation, inverse Compton scattering, bremsstrahlung, hadronic processes, radiation reprocessing)

VB, 5 sessions
16 – 24 Feb

3. **Accretion and ejection in relativistic sources**
    1. Accretion onto compact objects
    2. Outflows: jets and winds (general physical description)
    3. Flow dynamics (production, propagation, content, termination)
    4. Emission in relativistic outflows: electron-positron pairs
    5. Emission in relativistic outflows: protons and nuclei
    6. Radiation reprocessing

VB, 4 sessions
25 Feb – 4 Mar



Demonstration of software usage to model non-thermal radiation processes.

Program: 26 h: 24 h lectures, 1 h exam, 1 h presentation

MR, 2 sessions
9 – 10 Mar

4. Phenomenology of accreting sources with outflows



1. Observational tools (analysis and fundamental diagrams)
2. X-ray binary accretion modes
3. Disks and jets
4. Black holes at all scales: from X-ray binaries to AGNs

MR, 4 sessions
11 – 18 Mar

5. High-energy gamma-ray sources in the Universe

1. High-energy gamma-ray detectors and satellites
2. Imaging atmospheric Cherenkov telescopes
3. Galactic high-energy gamma-ray sources (pulsars, pulsar wind nebulae, supernova remnants, X-ray and gamma-ray binaries, etc.)
4. Extragalactic high-energy gamma-ray sources (AGNs, GRBs, EBL, etc.)
5. Fundamental physics at high-energy gamma rays (dark matter, Lorentz invariance, etc.)

Hands-on sessions with public software packages such as Naima.

Bibliography

AHARONIAN, F. A. Very high energy cosmic gamma radiation: a crucial window on the extreme universe. Singapore : World Scientific Publishing, cop. 2004.

CHARLES, PHILLIP A. ; SEWARD, FREDERICK D. Exploring the X-ray universe. Cambridge : Cambridge University Press, 1995.

LONGAIR, MALCOLM S. High energy astrophysics. Third ed. Cambridge : Cambridge University Press, 2010.

PACHOLCZYK, A. G. Radioastrofísica : procesos no térmicos en fuentes galácticas y extragalácticas . Barcelona : Reverté, DL 1979.

ROMERO, G.E.; PAREDES, J.M. Introducción a la astrofísica relativista. Textos docents 365. Publicacions i edicions Universitat de Barcelona (eBook: http://www.publicacions.ub.edu/ficha.aspx?cod=11608 or www.unebook.es)

Room A33M **Monday, Tuesday, Wednesday ~~and Thursday~~**

15:00-16:20 High Energy Astrophysics (09/02/2025 – 23/03/2025)
(Valentí Bosch, Marc Ribó)

Work required to the students:

- Students should **attend the lectures and participate actively** in class.
- Students must **take a written exam** at the end of the course.
- Students must **do a written work** on a topic of high-energy astrophysics proposed by the teachers.
- The work must also be **presented orally**.

Evaluation:

- Participation: 20%
- Written exam: 30%
- Written work: 20%
- Oral presentation of the work: 30%

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- ~~Students must **do a written work** ….~~
- Students must make an **oral presentation** of a topic of high-energy astrophysics proposed by the teachers.

Evaluation:

- Participation: 20%
- Written exam: 30% → 40%
- ~~Written work: 20%~~
- Oral presentation of the work: 30% → 40%

Proposed works and supervisors

Marc Ribó:

- M1. Supernovae at GeV-TeV
- M2. Pulsars or PWN at GeV-TeV
- M3. X-ray binaries at GeV-TeV
- M4. Gamma-ray binaries at GeV-TeV
- M5. Gamma-rays from star clusters
- M6. Blazars at GeV-TeV (options: EBL, neutrinos)
- M7. EM emission from NS-NS mergers

Valentí Bosch:

- V1. High-Energy Emission from Jet/Medium Interactions in AGN
- V2. High-Energy Emission from Microquasar Jets
- V3. Gamma-rays from Young Stellar Objects
- V4. Disk-Jet connections in Black Hole X-ray Binaries

Another topic of your interest, either observational or theoretical

Master Thesis proposals

Non-thermal emitters in Orion A
Advisors: Valentí Bosch and Gemma Busquet

Can compact accelerators inject PeV cosmic rays in the interstellar medium?
Advisor: Valentí Bosch

On the detectability of isolated black holes in molecular clouds
Advisor: Valentí Bosch

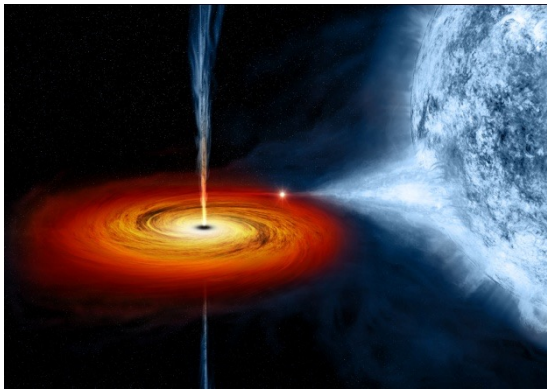
Transient sources with the LST-CTA
Advisor: Pol Bordas and Marc Ribó

Galactic sources with the Cherenkov Telescope Array
Advisor: Pol Bordas

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